



Research Article

The effects of fabrication atmosphere condition on the microstructural and mechanical properties of laser direct manufactured stainless steel 17-4 PH

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Abstract

The effects of atmosphere conditions on microstructural and mechanical properties of stainless steel 17-4PH components fabricated by laser direct manufacturing (LDM) were investigated through measurements on phase constitution, porosity, tensile strength, fracture morphology, hardness and evolution of substrate temperature. Results showed that the samples produced in air atmosphere condition possessed higher tensile strength and hardness for both as-deposited and heat-treated states than that in Ar chamber condition, due to dispersion strengthening effect of amorphous oxide particles and nitrogen solution strengthening as a result of higher content of oxygen and nitrogen. The temperature of substrate heat accumulation was higher in Ar chamber condition, leading to dramatically lower porosity and more reverse austenite, which also contributed to the lower strength and hardness.

Introduction

Stainless steel 17-4PH is a precipitation hardening martensitic stainless steel, which has advantages in high tensile and impact strength, fracture toughness and corrosion resistance [1]. Because of these attractive properties, it is widely used in aeronautic, astronautic, petroleum and nuclear industries [2].

Additive manufacturing (AM) is an emerging manufacturing method based on an incremental layer-by-layer deposition using energy of laser beam, electron beam, or electric arc, etc. [3]. In general, AM processes of metals can be divided into two categories according to the feeding methods, namely synchronous powder (or wire) feeding method (SPFM) and powder bed method (PBM) [4]. PBM is commonly used in selective laser sintering (SLS) [5] or selective laser melting (SLM) [6], while SPFM is usually used in LDM, which is a synonym of laser metal deposition (LMD) [7], direct metal deposition (DMD) [8], laser engineered net shaping (LENS) [9], etc., as shown in Fig. 1(a).

It is well known that PBM is basically used in an enclosed chamber, where the parts are manufactured in an inert gas environment, while SPFM can be used either in an sealed chamber filled with inert gas or in atmospheric environment with inert gas protection. Because of the unlimited dimensions, SPFM in atmospheric environment with inert gas protection is widely used for remanufacturing or additive manufacturing of large parts [10]. However, some foreign gases (O_2 , H_2O , N_2 , etc.) may enter the weld pool in spite of inert gas protection, which may affect the properties of the AM parts.

Although additive manufactured stainless steel 17-4PH components in the fabrication conditions either in an inert gas chamber or in atmospheric environment with inert gas protection have been widely studied separately, the effects of different fabrication atmospheres of the same AM system on the properties of 17-4PH components are seldomly reported in literature. In this work, a detailed research was conducted to investigate the properties of LDMed 17-4PH components in different atmosphere conditions. The results will be helpful to determine the optimal fabrication atmosphere condition for remanufacturing or additive manufacturing of large parts.

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Experimental

Experimental work was performed on a LDM-8060 laser direct manufacturing system, which consisted of a continuous wave Laserline fiber laser with maximum laser power of 3000 W, a 3-axis numerical control working table, a chamber with atmosphere control and an automatic powder feeder with coaxial nozzles. Chemical composition of stainless steel 17-4PH powder used in the present work is listed in Table 1. The size of the powder ranges from 45 to 105 μm . The dimension of the LDMed 17-4PH ...

Mechanical properties

The mechanical properties of as-deposited (AD) state and heat-treated (HT) state of the LDMed 17-4PH samples produced in air atmosphere condition with Ar protection and in Ar chamber condition, respectively, are listed in Table 3. For the same atmosphere condition, σ_b and $\sigma_s/\sigma_{0.2}$ increased after heat treatment, and for different atmosphere conditions, the strength was lower for samples in Ar chamber condition in both AD state and HT state. ...

Microhardness

Fig. 4 shows microhardness variations of the LDMed ...

Phases and mechanical properties

For different atmosphere conditions, the most apparent difference is the presence of foreign gases (e.g. H_2O , O_2 , N_2 , etc). From Table 3, one can find that the strength of AD state samples in Ar chamber condition was smaller than that in air atmosphere condition, which could attribute to the solution strengthening effect of nitrogen [14] and dispersion strengthening effect of oxide/carbide particles [10]. In fact, the chemical analysis results showed that the content of O and N in air ...

Conclusions

Stainless steel 17-4PH was fabricated in different atmosphere conditions using laser direct manufacturing technology. Some conclusions can be drawn as follows;

- (1) The fabrication atmosphere condition had great influence on the microstructure, phase constitution, porosity and mechanical properties of the LDMed 17-4PH components. ...

- (2) The substrate heat accumulation temperature was higher in Ar chamber condition than that in air atmosphere condition, due to the weaker heat dissipation caused by enclosed ...

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